Configural lesions in deep networks: analysis plan

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Purpose and status

This plan fixes every analysis that will be run on the hold-out set before any unit is selected or lesioned. It is frozen on Drive together with the hold-out fingerprint and timestamped in a GitHub repository; study models cannot be evaluated on the hold-out set until that freeze exists.

What is already done and fixed:

- **Stimuli.** The battery is frozen: Frankenstein with a 6 px horizontal cut, grid scramble with 12 px gaps, textured and silhouette renderings of every object.
- **Hold-out set.** 1500 COCO objects from images disjoint from the calibration set, fingerprint `91de2ec2b23333075e1b51e5157cd1975f1ba1ccff578cb13f56ba9cca8c86bf`, generated 15 October 2026. The local-only control (BagNet-33) replicated stimulus equivalence on it.
- **Models.** ResNet-50 (SWSL) is the primary model and CORnet-S the secondary, selected by a pre-specified rule on the localizer half of the calibration set.

The project has a single author, so every design decision below cites the literature it rests on rather than an individual's judgement. Analyses on the calibration set (Stages 3 to 4) are exploratory and may be iterated. Analyses on the hold-out set follow this document exactly. Any departure is recorded in the deviations log with its date and reason, and reported in the paper.

Background and hypotheses

The central prediction is that removing a small set of configuration-dependent units makes a network behave like an integrative agnosic: intact objects become as hard as scrambled ones, while local evidence survives.

In Sacks's account (1985), Dr P described a glove as a continuous surface folded in on itself and a rose as a red convoluted form with a green attachment. He reported parts and surface features accurately but could not name the whole. Integrative agnosia has the same profile: parts are perceived but not combined into wholes (Riddoch & Humphreys, 1987). Patients represent a few parts independently but fail when categorisation depends on part relations, and are near chance with overlapping figures (Behrmann & Williams,

2007). The integrative agnosic HJA lost the normal advantage of line drawings over silhouettes, consistent with internal contours fragmenting rather than supporting the whole (Lawson & Humphreys, 1999).

Standard CNNs are largely insensitive to object configuration (Baker & Elder, 2022), and our calibration data agree for Frankenstein stimuli. Grid scrambling, by contrast, costs the selected models far more than a local-only control. This gives a measurable configural component to remove.

The key outcome is configural reliance (CR): the scramble loss as a share of above-chance accuracy, defined in Outcome measures. A configural lesion should lower CR toward the local-only level. A random lesion of the same size should not.

ID	Hypothesis	Prediction	Family
H1	Both selected models rely on configuration on the hold-out set	CR excess over BagNet-33 above 0	Primary
H2	Configural units carry that reliance	CR falls more after the configural lesion than after size-matched and damage-matched random lesions	Primary
H3	The deficit dissociates from local processing	CR falls more after the configural lesion than after a lesion of local units of equal damage	Primary
H4	Local evidence survives, as in Dr P	Hue decoding and category decoding from scrambled images drop no more than after random lesions	Secondary
H5	Errors become local	Error confusions move toward BagNet-33's confusion pattern more than after random lesions	Secondary
H6	The line-drawing advantage shrinks, as in HJA	Edge-set minus silhouette-set accuracy falls more than after random lesions	Secondary
H7	Overlapping figures suffer most	Overlap cost of a multi-object read-out rises more than after random lesions	Secondary

Frankenstein effects are near zero in every model, so they carry no lesion hypothesis. They are reported descriptively as a model-human difference.

Data, stimuli and models

Units are selected on the calibration set and every confirmatory test runs on the hold-out set, which shares no photograph with it.

Set	Objects	Images	Use
Calibration, localizer half	880 (800 textured)	about 13,300	Model choice, unit scoring and selection
Calibration, test half	879 (799 textured)	about 13,300	Reliability of unit scores, lesion piloting
Hold-out	1500 textured, 15 categories	26,796	All confirmatory tests (H1 to H7)
Public reference sets	160 silhouettes, 160 edge drawings, 800 sketches, 1280 cue-conflict	2400	H6 and descriptive profile

Battery. Each object appears intact, with a 6 px horizontal cut (control), as two Frankenstein variants, with a 3 by 3 grid of 12 px gaps (control), and in three grid scrambles. Textured objects keep their photo pixels; silhouettes are binary. Hold-out Navon and overlapping-figure images use new seeds and hold-out objects.

Categories. The 16 model-vs-human categories are read out by averaging softmax over their ImageNet classes. The hold-out set has no knives, so hold-out means cover 15 categories, weighted equally.

Models. ResNet-50 (SWSL) is primary and CORnet-S secondary. They were selected from nine candidates by the pre-specified Stage 2 rule (scramble excess over BagNet-33: 0.190 [0.157, 0.224] and 0.103 [0.076, 0.130]). BagNet-33 is the local-only reference. The other seven candidates are reported descriptively and do not enter the lesion analyses.

Sites. Lesions target channels at four stage outputs per model: layer1 to layer4 in ResNet-50 and V1, V2, V4 and IT in CORnet-S, whose names map onto the ventral stream.

Outcome measures

The primary measure is configural reliance (CR), the scramble loss as a share of above-chance accuracy on the grid control; the absolute loss is reported alongside.

```
\mathrm{CR} = \frac{A_{\mathrm{grid}} - A_{\mathrm{scr}}}{A_{\mathrm{grid}} - 1/16}
```

Here A_grid is 16-way accuracy on the 12 px grid control and A_scr on the scrambles, averaged over an object's scrambles first. Scrambles that change less than 10% of the object's layout (1 minus IoU) are excluded. CR is 0 when scrambling costs nothing and 1 when it removes all above-chance accuracy. It removes the ceiling that made absolute losses depend on control accuracy (BagNet-33 and CLIP lose similar shares, 0.17 and 0.18, but very different absolute amounts).

CR became primary after the Stage 2 calibration run; the pre-specified absolute measure stays as a secondary outcome. A lesion's effect is the change it causes, ΔCR = CR(lesioned) minus CR(intact).

CR needs headroom to be meaningful. If a lesion leaves the lower 95% bound of A_grid below 0.125 (twice chance, the floor used since Stage 1), CR is not computed for that lesion. It is reported as floor and counts as not supporting any hypothesis that depends on it.

Measure	Definition	Used in
Absolute scramble loss	A_grid minus A_scr	H1 to H3, secondary
Damage	Drop in grid-control accuracy caused by a lesion	Matching control lesions
Hue decoding	Linear-probe accuracy for the object's dominant hue (6 bins, pixels with saturation above 0.2)	H4
Local category decoding	Linear-probe accuracy for category from scrambled-image features	H4
Error locality	Jensen-Shannon distance between the model's error distribution and BagNet-33's, over off-diagonal confusions	H5
Line-drawing advantage	Edge-set minus silhouette-set accuracy, paired over the same 160 images	H6
Overlap cost	Both-objects-correct rate of a multi-label probe, side by side minus small overlap	H7

All accuracies are 16-way and category-balanced. Frankenstein losses are computed the same way and reported descriptively.

Unit localisation

Channels are selected by their causal effect on configural reliance, not by response selectivity, because selectivity is a poor guide to what a unit does. Single-unit class selectivity predicts ablation damage poorly (Morcos et al., 2018) and is neither necessary nor sufficient for performance (Leavitt & Morcos, 2020). Dobs et al. found ranking filters by their effect on the loss more informative than ranking by selectivity, and lesioned the top 20% (Dobs et al., 2022).

Unit. A unit is one channel (feature map) at a site, ablated across all spatial positions, as for convolutional networks in Morcos et al. In CORnet-S the ablation acts on an area's output, so it removes that channel's signal to the next area after recurrence.

Configural importance (primary score). Each channel is zero-ablated alone and its effect measured on the localizer half, textured stimuli. P is the mean probability of the true category, which varies more smoothly than accuracy.

```
I_{\mathrm{conf}}(c) = \left[P_{\mathrm{grid}} - P_{\mathrm{grid}}^{(-
c)}\right] - \left[P_{\mathrm{scr}} - P_{\mathrm{scr}}^{(-c)}\right]
```

A positive score means the channel supports intact objects more than scrambled ones. The local score is the channel's support for scrambled objects, P_scr minus P_scr without c.

Amended (A1, 15 October 2026). The calibration pilot showed that this absolute difference favours channels that matter for every image, because intact objects start from a higher probability and so lose more in absolute terms. The primary score is now the channel's effect on configural reliance itself, computed on true-category probabilities: CR_p = 1 minus (P_scr minus 1/16) / (P_grid minus 1/16), and I_CR(c) = CR_p(intact) minus CR_p(with c ablated). Significance uses a one-sided, category-stratified bootstrap over objects (10,000 resamples) instead of the t-test, with the same Benjamini-Hochberg control and 1 to 40% bounds. Every rule that ranked by I_conf now ranks by I_CR. Full text: amendment_01.md in the preregistration repository.

Reliability. Scores are recomputed on the calibration test half. A site enters the lesion analyses only if the split-half Spearman correlation of channel scores, Spearman-Brown corrected, reaches 0.7. Failing sites are reported, not lesioned.

Sets. At each site, with n channels:

- **Configural set:** channels whose configural importance is reliably above 0, by a one-sided test across localizer objects with Benjamini-Hochberg FDR at q = 0.05, the same logic Prince et al. use for selective units. The set is bounded to 1 to 40% of the site's channels (the dose-response range); outside that range the top 1% or 40% is taken.

Dobs et al. give no rationale for their 20% and test no other size, and no biological lesion fraction maps onto channels: patients' lesions remove contiguous tissue, not a functionally chosen subset. The data therefore set the size.

- **Local set:** from channels with configural importance at or below the site median, the 20% with the highest local score.
- **Selective set (secondary):** channels whose spatially averaged response is higher for grid controls than scrambles, by a paired t-test across objects with Benjamini-Hochberg FDR at q = 0.05, as for category-selective units in Prince, Alvarez & Konkle (2024). Its overlap with the configural set (Jaccard index) tests whether selectivity and importance agree here.

Primary site. For each model, the reliable site whose configural set has the largest summed configural importance on the localizer half. The rule is fixed now and applied to calibration data only; all other reliable sites are reported as secondary.

Lesion protocol

The primary lesion zeroes the configural set (sized by the data, see Unit localisation) at each model's primary site and is compared with random lesions matched for size and for damage. All sets are fixed on the calibration set before the hold-out set is used.

Ablation. Channels are set to zero, the method used by Morcos et al., Dobs et al. and Prince et al. Every total ablation moves a network off its training distribution (Li & Janson, Optimal ablation for interpretability), so all primary contrasts are repeated with mean ablation (each channel replaced by its mean over calibration images). A result is called robust only if both methods agree in direction and significance.

Condition	How the channels are chosen	Role
Configural	FDR-significant (1 to 40% of channels) by configural importance	Target
Random, size-matched	100 random sets of the same size and site, seeded	H2 control, after Prince et al.
Random, damage-matched	From up to 10,000 random sets, the 100 whose damage is within 0.02 of the configural set's (else the 100 closest)	H2 control for total harm
Local	Top channels by local score, the set size chosen so its damage is closest to the configural set's	H3 contrast

Condition	How the channels are chosen	Role
Selective	FDR-selected configural-selective channels	Secondary

Damage is measured on the calibration localizer half, so matching never looks at hold-out results.

Secondary lesions.

- Dose-response at the primary site: 1, 2, 5, 10, 20 and 40% of channels, configural against 20 size-matched random sets per size.
- Every other reliable site, its configural set defined by the same rule, against 20 size-matched random sets.

Exploratory. Progressive lesioning with brief retraining, after Shah, Tran & Yamins (2024), tests whether configural reliance recovers or is replaced by local strategies. It is defined after the confirmatory analyses and reported as exploratory.

The networks are deterministic in evaluation mode; the only randomness is in the control sets, which use fixed seeds.

Agnosia signature tests

The secondary hypotheses test whether the lesioned network shows Dr P's profile: local evidence kept, wholes lost, errors driven by parts.

Probes. Read-outs are linear (L2-regularised logistic regression) on penultimate-layer features, because linear probes are the most selective in the comparison by Hewitt & Liang (2019). Probes are retrained for each lesion condition on calibration features, with the regularisation chosen by 5-fold cross-validation grouped by object, then tested on hold-out features. Each probe is paired with a control task, the same probe trained on labels shuffled at the object level; probe selectivity (task minus control accuracy) is reported.

- **H4, local evidence.** Two probes: dominant hue, and category trained and tested on scrambled images. Supported if each drop after the configural lesion exceeds the mean drop after damage-matched random lesions by less than 0.05 (upper confidence bound below 0.05). The same category probe on grid controls is reported to show the whole-object loss at the representation level.
- **H5, local errors.** On hold-out grid controls, the off-diagonal confusion distribution is compared with BagNet-33's by Jensen-Shannon distance. Supported if the distance falls more after the configural lesion than after damage-matched random lesions.

- **H6, line-drawing advantage.** Edge-set minus silhouette-set accuracy over the same 160 public images. Supported if it falls more after the configural lesion than after damage-matched random lesions. The reference sets are public, so this is the one test not run on hold-out images; the unlesioned values were seen in Stage 2.
- **H7, overlapping figures.** A 16-output multi-label linear probe, trained on calibration outlines and pairs, counts a trial correct when its top two outputs are the two objects. Supported if the overlap cost rises more after the configural lesion than after damage-matched random lesions. Patient S.M. was at chance on overlapping figures but better with the same shapes apart (Behrmann & Williams, 2007).

Exploratory. Navon letters (global and local letter probes), Frankenstein effects after lesions, and per-category profiles are reported without hypotheses.

Statistical analysis

Every test uses a category-stratified cluster bootstrap over objects (10,000 resamples), category-balanced means, and Holm correction within each hypothesis family.

Resampling. The object is the independent unit; its variants (scrambles, Frankenstein halves) are averaged first. Objects are resampled within category, so every resample keeps the category balance. Lesion conditions are resampled jointly, so contrasts between them are paired. CR is a ratio of means and is recomputed in each resample; intervals are percentile intervals.

Decision rules.

Hypothesis	Statistic	Supported if
H1	CR(model) minus CR(BagNet-33)	Lower 95% bound above 0
H2	ΔCR(configural) minus mean ΔCR(random), for size-matched and damage-matched controls	Both upper bounds below 0, and ΔCR(configural) below the 5th percentile of the 100 damage-matched random lesions
H3	ΔCR(configural) minus ΔCR(local)	Upper bound below 0
H4	Probe drop, configural minus damage-matched random	Upper bound below 0.05 (non-inferiority)
H5 to H7	Change after configural minus change after damage-matched random	Bound excludes 0 in the predicted direction

Multiplicity. The primary family is H1 to H3 in two models (6 tests); the secondary family is H4 to H7 in two models (8 tests). Two-sided bootstrap p-values are Holm-corrected within each family at a family-wise 0.05.

Margin. The 0.05 accuracy margin is the same bound that defined negligible local effects when the stimuli were validated, so "negligible" means one thing throughout. Equivalence and non-inferiority follow Lakens (2017), with the smallest effect of interest fixed in raw accuracy units.

Scope of claims. The primary model decides each headline result. A result is called general only if both models support it.

Sensitivity analyses. Mean ablation, the absolute scramble loss, source-weighted means, and all secondary sites. They are reported in full whether or not they agree with the primary analysis.

Failure and reporting

Every hypothesis is reported whatever its outcome, and each failure pattern already has a fixed interpretation.

Outcome	Interpretation	How it is reported
H1 fails for a model	Its configural reliance did not replicate beyond calibration	Lesion results for that model are reported as conditional on an unreplicated effect
No site passes reliability	Configural importance cannot be measured stably per channel	Reported as a negative result; no lesion claim for that model
H2 fails	Configural reliance is not concentrated in a sparse channel set at the tested sites	Reported as evidence for a distributed code, with the dose-response curve
H2 holds, H3 fails	The lesion harms recognition generally rather than integration specifically	No dissociation is claimed
Zero and mean ablation disagree	The effect depends on the ablation method	Called method-dependent, not robust
Some of H4 to H7 fail	The lesioned profile matches the syndrome only in part	The matching and non-matching features are listed side by side

Exploratory analyses are labelled as such and carry no confirmatory claim: progressive lesioning with retraining, unit characterisation (receptive fields, attribution maps), Navon probes, Frankenstein effects after lesions, per-category results, and the seven unselected models.

Design history and deviations

Six design changes were made before this plan, all on calibration data and all listed in the paper's Methods. Changes after the freeze are appended here, newest first; amendment A1 is the only one.

Date	Change	Reason	Data seen before the change
15 Oct 2026 (after freeze)	Amendment A1: unit score changed from absolute probability differences to the channel's effect on configural reliance (I_CR), tested by object bootstrap; the only revision to unit selection	The absolute score favoured generally important channels; the configural lesion raised CR in the pilot (ResNet-50 SWSL 0.542 to 0.690) and removed all recognition in CORnet-S	Stage 3 calibration scores, sets and pilot; hold-out untouched
15 Oct 2026	CLIP reloaded with its matching activation (QuickGELU) and Stage 2 rerun	Loader bug; selection unchanged	Stage 2 calibration results
15 Oct 2026	Configural reliance (CR) made primary; absolute loss kept as secondary	Absolute losses are capped by control accuracy	Stage 2 calibration results
15 Oct 2026	Four candidates added (adversarial and SWSL ResNet-50, ConvNeXt-T, CLIP)	Standard reviewer questions on robustness, data scale and modern models	None for these models
15 Oct 2026	Hold-out set generated and locked; knife has no hold-out objects	Independent confirmation; calibration used every eligible knife	BagNet controls only on hold-out

Date	Change	Reason	Data seen before the change
15 Oct 2026	Grid gap widened from 6 to 12 px by a pre-specified rule	BagNet-33 scramble drop 0.054 exceeded the 0.05 bound at 6 px	Stage 1 calibration results
15 Oct 2026	Textured renderings added to the silhouettes	Local-only controls were at chance on silhouettes	Stage 1 calibration results

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